Xiaochuan Li

M.Sc. Human and AI · University of Technology Nuremberg

xiaochuan.li@utn.de · github.com/xchuan-li · Nürnberg, Germany

EDUCATION

M.Sc., Human and AI 2025 — 2027

University of Technology Nuremberg (UTN)

Focus: causal interpretability through linguistic structure — testing whether a model's behavior is licensed by the genuine structure of a sentence, or a surface cue that merely correlates.

German language program 2024 — 2025

University of Marburg · intensive German for graduate study

B.A., Philosophy 2020 — 2024

Shenzhen University · College of Humanities

Senior thesis: *On Leibniz's Logical Theory* (advisor: Zang Yong)

Coursework: logic, history of philosophy, analytic philosophy, mathematical logic; electives in cognitive psychology and linguistics.

RESEARCH

Causal interpretability through linguistic structure

Thesis program · three method contributions

Certify whether behavior is licensed by genuine structure (Stable Is Not Grounded), localize where that structure is used (Isotrace), and quantify how much the model relies on it (Causal Inferential Yield) — developed and tested through linguistic structure.

Typological Grounding

Application frontier

Where the three methods meet linguistic typology — the same meaning re-encoded by each grammar; a test of structure versus surface.

Mini Causal Models · Latent Control States · Cross-lingual ProtoBias

Support

A validation sandbox and two extensions the program leans on — a mechanistic bridge into activation space and an applied multimodal stress test.

PAPERS IN PROGRESS

Stable Is Not Grounded. Paper 1: certification — whether the licensing structure is load-bearing.

Isotrace. Paper 2: localization — which reasoning hop is used, via path-encoded fictional names.

Causal Inferential Yield (CIY). Paper 3: quantification — how much genuine structure the model relies on.

EXPERIENCE

IT Student Assistant
Apr 2026 — present

University of Technology Nuremberg

Hardware maintenance and general IT support for staff and teaching labs.

Editor & public-facing writer
2020 — 2024

Student magazine, Shenzhen University · personal essays (WeChat)

Deputy editor-in-chief of a student magazine; author of personal philosophy essays (~10,000 reads) on determinism, free will, and causality — the questions that became my research.

SKILLS

Consolidating the technical stack my research program needs — controlled linguistic data generation, transformer-based model evaluation, intervention-based causal testing, and reproducible experiment pipelines.

Programming & data. Python (NumPy, pandas, scikit-learn), Git, LaTeX.

ML / NLP tooling. Building working proficiency with PyTorch and HuggingFace Transformers — model inference, dataset pipelines, and LoRA fine-tuning.

Causal & interpretability methods. Controlled do-interventions, causal graphs, and shortcut-severing audits; developing behavioral path-tracing.

Linguistic evaluation. Designing controlled stimuli for morphology, agreement, dependency structure, and word order — and for multilingual / typological variation.

LANGUAGES

Mandarin Chinese (native) · German (C1) · English (C1) — spanning isolating, fusional, and analytic types. Learning Japanese, to extend the range with an agglutinative, head-final system — the typological span my linguistic work draws on.

INTERESTS

Causal interpretability through linguistic structure — whether a model represents the genuine structure of language or merely its surface. The boundary between behavioral and mechanistic methods. Philosophy of measurement and of language in ML. World-model approaches (LeCun, Pearl) and the limits of pure-LLM paradigms.